

Exploratory Data Analysis (EDA) and Machine Learning on Agricultural Yield Dataset

Part A: Understanding the Dataset

Q1. Dataset Overview

Number of Rows and Columns

- **Rows:** 1500
- **Columns:** 8

Column Names

1. rainfall_mm
2. temperature_c
3. fertilizer_kg
4. irrigation_hours
5. soil_ph
6. crop_type
7. soil_type
8. yield_ton_per_hectare

First 10 Records

rainfall_mm	temperature_c	fertilizer_kg	irrigation_hours	soil_ph	crop_type	soil_type	yield_ton_per_hectare
588.6	18.6	...	...	...	...	Loamy	5.92
772.8	34.6	...	...	...	...	Sandy	6.24
970.9	36.3	...	...	...	...	Sandy	4.77
611.7	19.0	...	...	...	...	Loamy	4.04

696.1	29.6	...	...	...	...	Clay	5.17
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(Use `df.head(10)` to display all 10 rows in your notebook.)

Q2. Data Types and Missing Values

Data Types

Column	Data Type
rainfall_mm	float64
temperature_c	float64
fertilizer_kg	float64
irrigation_hours	float64
soil_ph	float64
crop_type	object
soil_type	object
yield_ton_per_hectare	float64

Missing Values

Column	Missing Values
rainfall_mm	0
temperature_c	0
fertilizer_kg	0
irrigation_hours	0
soil_ph	0
crop_type	0

soil_type 0
yield_ton_per_hectare 0

Conclusion

- No missing values are present.
- The dataset is clean and ready for analysis.

Q3. Descriptive Statistics

Summary Statistics

Feature	Mean	Std Dev
rainfall_mm	754.05	255.10
temperature_c	27.75	5.76
fertilizer_kg	149.98 (approx.)	—
irrigation_hours	—	—
soil_ph	6.76	0.72
yield_ton_per_hectare	5.03	0.97

Answers

Feature with highest mean value

- rainfall_mm (754.05)

Feature with highest standard deviation

- rainfall_mm (255.10)

Part B: Exploratory Data Analysis (EDA)

Q4. Distribution Analysis

Create histograms for:

```
df['rainfall_mm'].hist()  
df['temperature_c'].hist()  
df['fertilizer_kg'].hist()  
df['yield_ton_per_hectare'].hist()
```

Observations

rainfall_mm

1. Values are distributed between 300 mm and 1200 mm.
2. Distribution appears approximately uniform.
3. No significant outliers are visible.

temperature_c

1. Temperatures range from 18°C to 38°C.
2. Distribution is fairly symmetric.
3. No extreme values observed.

fertilizer_kg

1. Fertilizer usage is spread across the entire range.
2. Distribution appears balanced.
3. No obvious skewness.

yield_ton_per_hectare

1. Yield values are concentrated around 5 tons/hectare.
2. Distribution resembles a bell shape.
3. Very few extreme observations.

Q5. Crop Type Analysis

Frequency of Each Crop

Crop	Count
Cotton	311

Soybean	306
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Wheat	303
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Rice	293
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Maize	287
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Count Plot

```
sns.countplot(x='crop_type', data=df)  
plt.show()
```

Most Frequent Crop

Cotton (311 records)

Q6. Soil Type Analysis

Frequency of Soil Types

Soil Type	Count
Clay	534
Sandy	492
Loamy	474

Count Plot

```
sns.countplot(x='soil_type', data=df)  
plt.show()
```

Most Common Soil Type

Clay (534 records)

Q7. Yield Distribution

Histogram

```
sns.histplot(df['yield_ton_per_hectare'], kde=True)
```

```
plt.show()
```

Answers

Is the distribution approximately normal?

- Yes, it is approximately bell-shaped and close to a normal distribution.

Any noticeable outliers?

- No major outliers are visible.

Q8. Scatter Plot Analysis

Rainfall vs Yield

```
sns.scatterplot(  
    x='rainfall_mm',  
    y='yield_ton_per_hectare',  
    data=df  
)
```

Fertilizer vs Yield

```
sns.scatterplot(  
    x='fertilizer_kg',  
    y='yield_ton_per_hectare',  
    data=df  
)
```

Answer

Rainfall shows a stronger positive relationship with crop yield compared to fertilizer usage.

Q9. Correlation Analysis

Correlation with Yield

Feature	Correlation
rainfall_mm	0.554
irrigation_hours	0.543

fertilizer_kg	0.278
soil_ph	0.024
temperature_c	-0.023

Top 3 Features Correlated with Yield

1. **rainfall_mm (0.554)**
2. **irrigation_hours (0.543)**
3. **fertilizer_kg (0.278)**

Heatmap

```
sns.heatmap(  
    df.corr(numeric_only=True),  
    annot=True,  
    cmap='coolwarm'  
)  
plt.show()
```

Q10. Group-Based Analysis

Average Yield by Crop Type

Crop Type	Avg Yield
Rice	5.495
Soybean	5.173
Wheat	4.989
Maize	4.897
Cotton	4.607

Highest Average Yield Crop

Rice (5.495 tons/hectare)

Average Yield by Soil Type

Soil Type	Avg Yield
Loamy	5.367
Clay	5.134
Sandy	4.589

Highest Average Yield Soil

Loamy Soil (5.367 tons/hectare)

Part C: Data Preparation

Q11. Feature Encoding

Categorical Columns

1. crop_type
2. soil_type

One-Hot Encoding

```
df_encoded = pd.get_dummies(  
    df,  
    columns=['crop_type','soil_type']  
)
```

First Five Rows

The transformed dataset contains:

- Original numerical columns
- crop_type_Cotton
- crop_type_Maize
- crop_type_Rice
- crop_type_Soybean
- crop_type_Wheat
- soil_type_Clay
- soil_type_Loamy

- soil_type_Sandy

Total columns after encoding = 14

Q12. Feature Selection

Target Variable (y)

```
y = df_encoded['yield_ton_per_hectare']
```

Input Features (X)

```
X = df_encoded.drop(
    'yield_ton_per_hectare',
    axis=1
)
```

Target Column

yield_ton_per_hectare

Part D: Machine Learning

Q13. Train-Test Split

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

Shapes

Dataset	Shape
X_train	(1200, 13)

X_test (300, 13)

y_train (1200,)

y_test (300,)

Q14. Linear Regression Model

Training

```
from sklearn.linear_model import LinearRegression
```

```
model = LinearRegression()  
model.fit(X_train, y_train)
```

Intercept

1.9111

Top Positive Coefficients

Feature	Coefficient
crop_type_Rice	0.477
soil_type_Loamy	0.366
irrigation_hours	0.198
crop_type_Soybean	0.095
soil_type_Clay	0.063

Feature with Highest Positive Coefficient

crop_type_Rice (0.477)

This indicates that, keeping all other variables constant, Rice cultivation contributes the most positively to yield prediction in the linear regression model.